



**AA STRUCTURAL
ENGINEERS**

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Structural Specification

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General

1. Do not scale from the drawings. The Contractor is to check all dimensions on site before carrying out works.
2. This specification is to be read in conjunction with the Architect's and all other Consultants' drawings and specifications, which should be used to verify layout, setting out, finishes etc. Any discrepancies are to be brought to the attention of the Architect prior to construction.
3. Where notes in the specification differ from the drawings and details, clarification should be sought from the Engineer.
4. The Contractor is to inform the Architect and Structural Engineer if the existing fabric, including foundations, is opened up and found to be inadequate, unsuitable to support the proposed works, or at variance from the details shown on the drawings.
5. Items noted on the drawings "to be confirmed on site" are to be exposed by the Contractor for inspection by the Structural Engineer at the earliest opportunity.
6. Do not cut any holes or chases through any structural members without first obtaining the written consent of the Structural Engineer.
7. The Client must appoint a principal designer and they must inform the Health and Safety Executive of the works by filling in and submitting an F10 form prior to starting work on site where a project is notifiable.
8. The Contractor must ensure that the Contract Administrator has agreed all necessary Party Wall notices prior to carrying out works under, on or adjacent to a Party Wall.
9. The Contractor is to ensure that the Building Control Officer and Structural Engineer are notified to carry out inspections of any structural work prior to covering up with finishes.
10. This specification and design uses Eurocodes which is cited in Building Regulations part A & is the current best practice. All Contractor design using loads from our design is to be similarly compliant with Eurocodes unless agreed otherwise in writing.

Temporary Works

1. The Contractor is entirely responsible for maintaining the stability of all existing buildings and structures, within and adjacent to the works, and of all the works from the date of possession of the site until practical completion of the works.
2. The Contractor shall design, install and maintain all necessary temporary works and shall submit proposals for the temporary supports and sequence of construction for the works, to the Structural Engineer and Contract Administrator at least 10 working days prior to starting on site. These proposals shall be supported by design calculations unless agreed otherwise by the Structural Engineer in writing.

Stability

1. The Contractor is to accept full responsibility for the stability and structural integrity of the works during the Contract and provide temporary support as necessary. They shall also prevent overloading of any completed, partially completed or existing elements.

Tolerances

1. All tolerances are to be agreed with the Architect and in accordance with their drawings and specifications, and the Contractor will be responsible for ensuring that sufficient tolerances are provided and integrated throughout all elements of the works.
2. The Contractor is to take account of tolerances detailed elsewhere in all relevant project drawings and specifications when complying with the above clause.
3. Unless otherwise indicated on the drawings and specifications provided by the Architect or Engineer the setting out dimensions and levels of the finished structural works shall be within the maximum tolerances given below:

Description	Maximum Tolerance
All dimensions of 3m and over	+/- 5mm
All dimensions less than 3m	+/- 3mm

Materials and Workmanship

1. All articles, materials and goods shall be new and of good quality, suitable for the required purpose and shall conform to the appropriate Eurocode, British Standard (if still applicable), or other applicable quality standard where such exists. Where references to the above are made it shall be inferred that the latest edition applies, together with subsequent amendments, unless otherwise specified.

Masonry

1. Workmanship is to comply generally with BS EN 1996-1-1, 2 & 3. Brickwork to be to BS EN 772-3 & 7 and BS EN 771-1. Blockwork to be to BS 6073-2.
2. New brickwork above the damp proof course (d.p.c.) is to be minimum class 3 set in 1:1:6 (cement:lime:sand) mortar, unless noted otherwise on the drawings.
3. New blockwork is to be minimum strength 7.0 N/mm² set in 1:1:6 (cement:lime:sand) mortar, unless noted otherwise on the drawings.
4. Brickwork and blockwork are to be laid properly bonded as agreed with the Architect and fully bonded into existing masonry. All junctions in structural walls are to be fully tooth bonded unless agreed otherwise.
5. All masonry below the d.p.c. is to be set in 1:3 (Cement:Sand with plasticiser) mortar with sulphate resisting cement.
6. New brickwork below the d.p.c. is to be Class B engineering bricks.
7. New blockwork below the d.p.c. is to be specified as suitable for such use by the manufacturer, and of minimum strength 7.0 N/mm², or as noted otherwise on the drawings.
8. Do not use frozen materials or lay masonry when the ambient air temperature is at or below 3°C and falling or unless it is at least 1°C and rising.
9. Unless noted otherwise on drawings or Architects specification, Cavity wall ties shall be Ancon Staifix RT2 stainless steel ties (or similar equivalent) to BS EN 845-1:2003 spaced at 450mm centres vertically, 900mm centres horizontally staggered, and at 225mm centres vertically at 225mm from all openings, corners and reveals. Minimum embedment to be 62.5mm into each masonry leaf.
10. Wall ties elsewhere are to be, to BS EN 845-1:2003, as noted on the drawings. Minimum embedment to be 62.5mm into each masonry leaf.
11. Bricks and blocks shall not be stored on any floor without first obtaining written consent from the Engineer. The Contractor shall ensure that the loadings imposed on the permanent works by the storage of materials does not overstress any part of the permanent works or cause excessive deflection.
12. In dry weather, bricks are to be soaked in water before being laid, and tops of walls to be raised are to be similarly soaked before work is recommenced.
13. Brickwork and blockwork is to be carried up in a uniform manner and is to be raked back and not toothed up, no section rising more than one metre above the remainder. Brickwork built with standard 65mm bricks shall rise at the rate of four courses to 300mm. No more than sixteen courses shall be built in a day without prior permission of the Engineer.
14. Crack control brick reinforcement is to be provided over doors, over and under windows and at changes in profile (e.g. where the building steps from two storeys to one storey), as follows: 2 layers of Bekeart Bricktor (or similar equivalent) in the two bed joint immediately adjacent to the opening. To extend 600mm beyond the opening on both sides and 600mm either side of the change in profile.
15. Vertical movement joints should be provided in masonry walls built with cementitious mortar to minimise the risk of major cracking, as shown in the following table:

Material	Joint Width (mm)	Normal Spacing
Clay brick	16	12m (15m maximum)
Calcium silicate brick	10	7.5 to 9m
Concrete block and brick	10	6m
Any masonry parapet wall	10	Half the above spacing and 1.5m from corners (double the frequency)

The spacing of the first movement joint from a return should not be more than half of the above dimension.

16. Provide flat straight stainless steel ties within the joint at 225mm vertical centres de-bonded with sleeves over one half. Joints to be filled with suitable compressible material with minimum 10mm deep weather proof sealant to the external leaf. In cavity walls, provide cavity wall ties (as clause 9), at 225mm centres vertically within 225mm of either side of the joint. Positions of joints to be agreed with the Architect prior to construction.
17. Steel columns, posts and proprietary windposts to be tied to internal block leaf within cavity walls using Halfen HTS framing cramps at 225mm vertical centres, or similar equivalent product, fixed to steel in accordance with manufacturers specification.
18. Proprietary wall starter systems such as Furfix (or similar equivalent) may be used to tie new masonry extensions to existing masonry in locations where approved by the Structural Engineer.
19. Use proprietary head restraints as detailed by Halfen or Ancon (or similar equivalent) to tie tops of internal block walls to the underside of concrete floor slabs.
20. All masonry walls to be restrained by timber floors and roofs with lateral restraint straps at 1200mm c/c as noted in the Timber specification section.

Timber

1. TIMBER PROCUREMENT

Timber (including timber for wood based products): obtained from well-managed forests and/or plantations in accordance with:

The laws governing forest management in the producer country or countries.

International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).

Documentation: provide either:

Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied.

Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood based products.

2. New timber in the works is to be selected structural timber not inferior to European Redwood/Whitewood grade C24 to BS EN 338:2009, unless noted otherwise on the drawings.
3. Timber that in the opinion of the Engineer is inferior in quality or condition or is not suitable for the requirements of the work is not to be used.
4. All existing timber is to be inspected for signs of damage, decay or infestation at the beginning of the project by a specialist. Refer to specialist's report for all information in connection with timber treatment or replacement.
5. New timber in the works is to be vacuum impregnated with preservative to BS EN 1995-1-1 and BS EN 351-1 and the manufacturer's recommendations. Cut ends are to be thoroughly treated with brush applied coats of appropriate preservative before fixing. All preservatives are to be to the Architect's approval.
6. Timber shall be stored under cover, clear of the ground and protected from the weather.

7. Structural timbers may only be drilled or cut for services with the approval of the Structural Engineer. Drill holes and notches in joists to be in accordance with NHBC Standards Chapter 6.4-S9 and BS EN 1995-1-1.
8. Sizes of new structural timbers noted on the drawings are sawn basic sizes.
9. All proprietary joist hangers, straps, connectors etc to be provided by Simpson Strong-tie or Expamet, or similar approved, and shall be fit for purpose.
10. All timber connectors, screws, nails, joist hangers, steel straps etc are to be galvanised or sherardised. All such items are to be fixed in accordance with the manufacturer's recommendations, unless shown otherwise on the drawings.
11. All doubled or tripled up timbers shall be fixed together with M12 bolts or threaded rods at 600mm centres, using double toothplate connectors between timbers, and 50mm square steel washers (3mm thick), unless shown otherwise on the drawings.
12. Unless otherwise specified, securely fix full depth solid timber noggins between joists as follows:
 - a. Joist spans of up to 3.5m: one row at third spans and at the bearings.
 - b. Joist spans over 3.5m: rows at quarter spans, mid span and at the bearings.
 - c. At positions of joints in ply sheeting.
13. Wall plates for roofs are to be tied down using 1200mm long 30 x 2.5mm galvanised mild steel straps at maximum 1200mm centres with 100mm bob end. Straps are to be nailed to the top plate and plugged and screwed to the internal face of the wall with a minimum of 3 No.12 wood screws 50mm long.
14. Lateral restraint straps to the perimeter of all floors, roofs and where otherwise noted on the drawings are to be minimum 700mm long 30 x 5mm galvanised mild steel straps at maximum 1200mm centres with 100mm bob end. Straps perpendicular to joists are to be fixed to the tops of three joists over solid blocking infill and notched in a maximum of 6mm. Straps parallel to joists are to be fixed to the side face of the joist. Straps to be fixed with a minimum of 3 (No.12) wood screws 50mm long.
15. Where sections of floor or roof are separated by a steel beam install 1250mm long 30 x 5mm straps at 1200mm c/c for continuity.
16. Where lateral restraint straps for floors are to be tied into existing 215mm thick (min) solid brick walls, cast bobbled end into a 225x100x150dp mass concrete anchor block into pocket and dry packed at top with 3:1 sharp sand cement well rammed in.

Steelwork

1. The design, fabrication and erection of the structural steelwork is to be in accordance with the current version of BS EN 1993-1-1 and BS EN 1090-2 and the latest edition of the National Structural Steelwork Specification (NSSS and NSSS CE Marking version) for Building Construction, and all clauses, including appendices are deemed to be part of this specification.
2. All structural steel sections are to be Grade S355 (unless agreed otherwise or noted otherwise on the drawings) to the applicable code from the following list; BS 4-1:2005, BS EN 10210-2:2006, BS EN 10025-2:2004.
3. The steel connections are to be designed and detailed by the Engineer drawings will be issued for construction.
4. The steelwork fabricator is to obtain dimensions from site. Setting out dimensions are to be obtained from the Architect's drawings.
5. Shop fabrication drawings showing layout, connections and fixing details, and connection calculations where requested, are to be submitted to the Engineer and Architect for comment at least two weeks before any fabrication is carried out.
6. All bolts are to be grade 8.8 Black Bolts (unless noted otherwise on the drawings) to BS 4190. Designed steel to steel connections to use a minimum of 2No. M16 bolts unless noted otherwise in the details.
7. All welding is to comply with BS EN 1011 Parts 1 & 2. Site welding shall not be permitted except with the written approval of the Structural Engineer. Where permitted, all site welding to be 100% tested (unless agreed otherwise with Engineer) in accordance with the National Structural Steelwork Specification (NSSS). All site weld test reports

to be submitted to the Structural Engineer at least 10 working days prior to the covering of the site welded areas with permanent finishes.

8. All welds are to be 6mm fillet welds or full strength butt welds unless noted otherwise on the drawings. Carry out additional weld testing in accordance with the National Structural Steelwork Specification for Building Construction on any critical welds specified by the Structural Engineer.
9. Unless noted otherwise, ends of all steelwork built into brickwork are to be concrete encased. Min 50mm concrete cover.
10. Unless noted otherwise, all steelwork in contact with the ground to be wrapped in D49 mesh and encased in concrete, 50mm minimum cover.
11. Unless noted otherwise, steel columns within box frames installed to form openings in existing masonry walls are to be bolted to the existing masonry using M12 resin anchors at 600mm c/c staggered vertically (100mm min embedment unless noted otherwise). Use Hilti HIT-HY-70 resin system (or similar equivalent). Masonry reveals must be made good with new masonry fully tooth bonded into existing prior to fixing.
12. New steel beams or frames below existing walls are to be installed on the centreline of the wall unless noted otherwise on the drawings. Dry pack to be 3:1 sharp sand cement well rammed in.
13. Steel baseplates to have grout access and vent hole, and must be levelled with steel shims positioned symmetrically around baseplate perimeter. Do not use a central pack.
14. Column bases to be grouted in accordance with BS EN 1090-2. For spaces less than 25mm – obtain instructions. For spaces 25-50mm use 1:1 cement:sand mortar, just fluid enough to pour. Tamped well as filling proceeds. For spaces 50-80mm deep use 2:1 cement:sand mortar, just damp, tamped well against fixed supports. Portland Cement to BS EN 197-1 – CEM 1 42.5 or 52.5. Sand to BS EN 12620, grade 0/4 or 0/2 (MP). Additives not required.

Steelwork Paint Systems for Corrosion and Fire Protection

1. All painting shall be carried out in accordance with BS EN ISO 12944, BS EN 1090-2 and the paint manufacturer's instructions. Paint systems as specified below are by Sherwin-Williams Protective & Marine Coatings (contact: 01204 556 457 or sales.uk@sherwin.com). Contractor to provide details of any alternative manufacturers and paint specifications for approval prior to fabrication).

For a warm, dry building with no risk of condensation on steelwork i.e. internal steelwork concealed and fire protected with plaster finishes:

- After preparation by blast cleaning to Sa 2½ to BS EN ISO 8501-1 (Welds/edges/areas with surface imperfections: To BS EN ISO 8501-3, preparation grade), all surfaces, which shall be dry, shall be painted with one coat of zinc phosphate primer (75 microns dry film thickness (dft) Sherwin-Williams Protective & Marine Coatings Macropoxy C400V3). This coat should be applied in the works with any subsequent damage made good on site.

For steelwork within the perimeter wall of a building, such as within the inner leaf of a cavity wall but not in contact with the external leaf:

- After preparation by blast cleaning to Sa 2½ to BS EN ISO 8501-1 (Welds/edges/areas with surface imperfections: To BS EN ISO 8501-3, preparation grade), all surfaces, which shall be dry, shall be painted with one coat of zinc phosphate primer (125 microns dry film thickness (dft) Sherwin-Williams Protective & Marine Coatings Macropoxy C400V3). This coat should be applied in the works with any subsequent damage made good on site.

For steelwork in contact with the external leaf:

- Apply locally to contact area, 400 microns dft of Sherwin-Williams Protective & Marine Coatings Macropoxy M922 (shop applied spray) or 400 microns dft of Sherwin-Williams Protective & Marine Coatings Macropoxy M922m (mastic version applied by brush on site).
- these paints can be applied directly to a blast cleaned substrate.

As an alternative to using a bituminous paint product to coat ends of steel beams in contact with external walls etc, use 2 coats of 125 microns dft of Sherwin-Williams Protective & Marine Coatings Macropoxy L524, over 75 microns dft of Sherwin-Williams Protective & Marine Coatings Macropoxy C400V3.

2. Fire protection to all steelwork to suit the Architect's details. Any structural steel elements not suitably protected by the Architect's finishes or to be left exposed in the permanent condition are to be protected using intumescent paint system and dft loading schedule as specified by Sherwin-Williams Protective & Marine Coatings to suit the steel section size, and level of fire protection required.
3. Where intumescent paint is applied, provide the Engineer and Architect with a copy of the intumescent loading schedule as specified by the supplier, and a certificate to confirm the actual dry film thickness measured. This can be carried out by an independent specialist or self-certified by the Contractor.
4. Where indicated on the drawings the steelwork and fixings shall be hot dipped galvanised to BS EN ISO 1461 in order to give a uniform zinc deposit of at least 100 microns.
5. Galvanised steelwork to be for internal wet or external conditions should be treated as follows:
 - Flash/sweep blast using non-metallic abrasive within 20-30 microns typical or clean & degrease with Degreaser W500, fresh water rinse & allow to dry before applying Mordant Wash L703.
 - Apply one coat of primer - Sherwin-Williams Protective & Marine Coatings Macropoxy K267 100 microns dft.
 - Apply finish coat of Sherwin-Williams Protective & Marine Coatings Acrolon C137V2, 50 micron dft which comes in all RAL colours

If the finish is to be a light or dark metallic grey (micaceous iron oxide) the primer – Sherwin-Williams Protective & Marine Coatings Macropoxy K267 should be decreased to 75 microns dft and a finish coat of Sherwin-Williams Protective & Marine Coatings Macropoxy K267 75 microns dft should be applied.
6. Where steelwork is galvanised, in order to minimise problems with Liquid Metal Assisted Cracking (LMAC), the following restrictions should be adhered to for all connections designed by the Contractor:

Partial end plates – Avoid: use full end plates or bolted cleat connections.

Part depth stiffeners – Avoid: use full depth stiffeners welded with intermittent fillet welds.

Use intermittent fillet welds for attachment of brackets.

Prior to erection or application of other coatings, all galvanised structural steelwork is to be visually inspected for cracks or indications of LMAC cracking. Inspection is to be carried out by a suitably qualified person trained and competent in visual inspection for LMAC. Where suspected LMAC defects are identified inform the Engineer immediately.

Lintels

1. Prestressed concrete lintels to BS EN 845-2 by Supreme Concrete Ltd, Coppingford Hall, Coppingford Rd, Sawtry, Huntingdon PE28 5GP, Tel. 01487 833 300. Sizes and types as indicated on the drawings. End bearing lengths are to be at least 150mm for spans up to 1.5m, and 225mm for spans up to 2m, unless noted otherwise on the drawings.
2. Galvanized steel lintels are to be to BS EN 10346 by Catnic, Pontypandy Industrial Estate, Caerphilly, Mid Glamorgan CF83 3GL, Telephone 029 2033 7900. Sizes and types as indicated on the drawings. End bearing lengths are to be at least 150mm for spans up to 1.5m, and 225mm for spans up to 2m, unless noted otherwise on the drawings.
3. The Contractor shall submit details to the Engineer and Contract Administrator for any alternative lintel type or manufacturer and obtain the Contract Administrator's written approval, prior to commencement of the work.